On the Rankin Constants of Lattices

F. Oggier

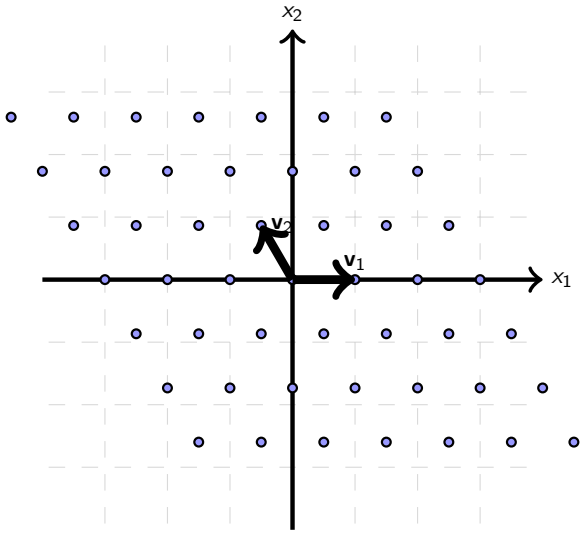
WCC 2026

Lattices

Coding Theory **Cryptography**

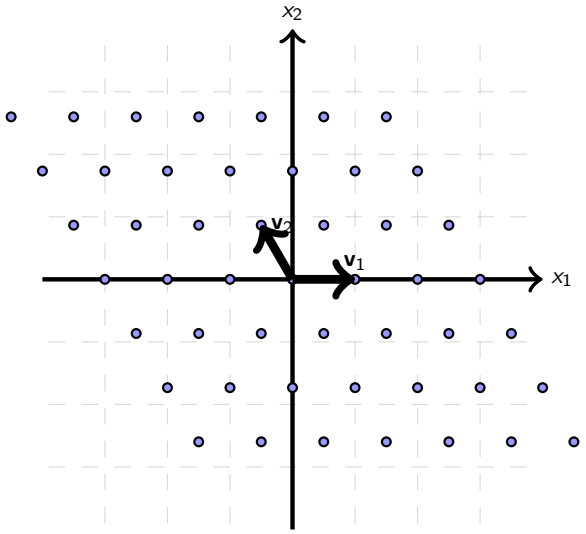
Suppose $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are linearly independent vectors in $\mathbb{R}^n$ $(m \leq n)$. A lattice $L \subset \mathbb{R}^n$, generated by $\mathbf{v}_1, \ldots, \mathbf{v}_m$, is the additive subgroup of $\mathbb{R}^n$ generated by $\mathbf{v}_1, \ldots, \mathbf{v}_m$:

$$L = \{\mathbf{u}M,\ \mathbf{u} \in \mathbb{Z}^m\},\ M = \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_m \end{bmatrix},\ G = MM^T.$$



$$M = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix},\ G = \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}.$$

The matrix M containing the basis vectors is called a generator matrix, the matrix $G = MM^T$ is called a Gram matrix (it contains the "geometry" of the lattice, that is the norms of the basis vectors on its diagonal, and their inner product or angle elsewhere).



$$M = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, \; G = \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}.$$

✓ **Lattices**

Coding Theory

Cryptography

- A vector $\mathbf{x} = (x_1, \ldots, x_k)$ of information symbols is encoded linearly into a codeword $\mathbf{c} = (c_1, \ldots, c_n)$:

$$\mathbf{x} = (x_1, \ldots, x_k) \mapsto \mathbf{c} = (c_1, \ldots, c_n), n \geq k,$$

  both have coefficients in a given alphabet, typically a finite field $\mathbb{F}_q$ or the finite ring $\mathbb{Z}_q = \{0, 1, \ldots, q-1\}$ of integers modulo q (if $q = p$ is a prime, $\mathbb{Z}_p = \mathbb{F}_p$).
- The set of all codewords, denoted by C, is called a linear code of length n.

If the alphabet is $\mathbb{F}_q$, C forms a linear subspace of $\mathbb{F}_q^n$. It thus has a dimension, namely k, and we say that we consider an (n, k) linear code C over $\mathbb{F}_q$.

- Since an (n, k) linear code C is a subspace, it has a basis, containing k vectors, that we stack as rows of a $k \times n$ matrix G, called a generator matrix for C.
- There is a unique generator matrix of the form $G = [\mathbf{I}_k | A]$ where $\mathbf{I}_k$ is the identity matrix. The code is said to be in systematic form.

$$\underbrace{(x_1, \ldots, x_k)}_{\text{information data}} \underbrace{\begin{bmatrix} & a_{11} & a_{1,n-k} \\ \mathbf{I}_k & \vdots & \vdots \\ & a_{k,1} & a_{k,n-k} \end{bmatrix}}_{\text{generator matrix } G} = \underbrace{(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)}_{\text{codeword}}$$

✓ **Lattices**

✓ **Coding Theory**

Cryptography

Let

$$\rho : \mathbb{Z} \to \mathbb{Z}_q = \{0, 1, \ldots, q-1\},\ x \mapsto x \pmod{q},$$

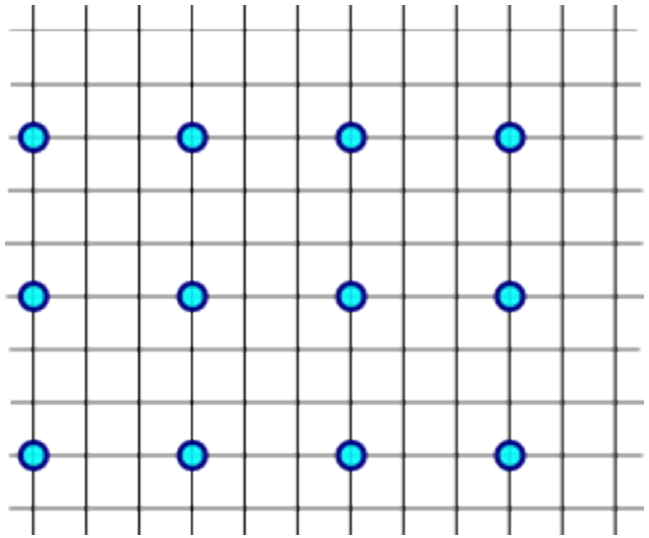
be the map of reduction modulo q. Given $a \pmod{q}$, its pre-image $\rho^{-1}(a)$ is the set of integers that are mapped to a by ρ, that is $\rho^{-1}(a) = \{a + bq,\ b \in \mathbb{Z}\}$.

The pre-image $\rho^{-1}(a) \subset \mathbb{Z}$ of $a \pmod{5}$.

Consider

$$\rho : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}_q \times \mathbb{Z}_q,\ (x_1, x_2) \mapsto (x_1 \pmod q, x_2 \pmod q).$$

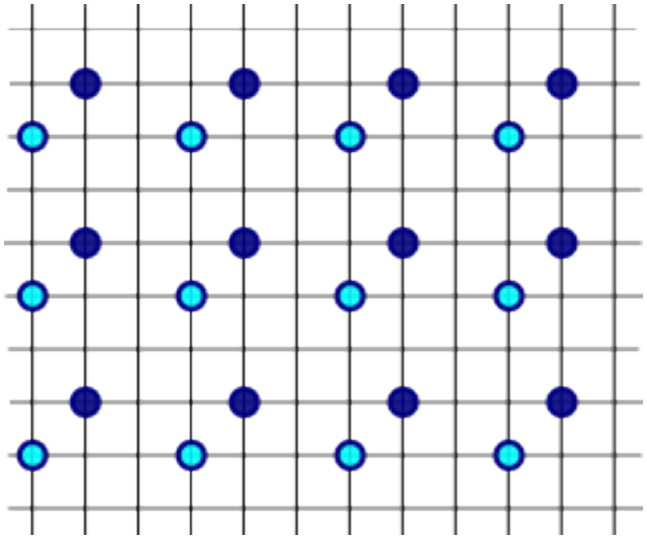
The pre-image $\rho^{-1}((a_1, a_2))$ is now the set of 2-dimensional vectors with integer entries that is mapped to (a_1, a_2) by ρ.



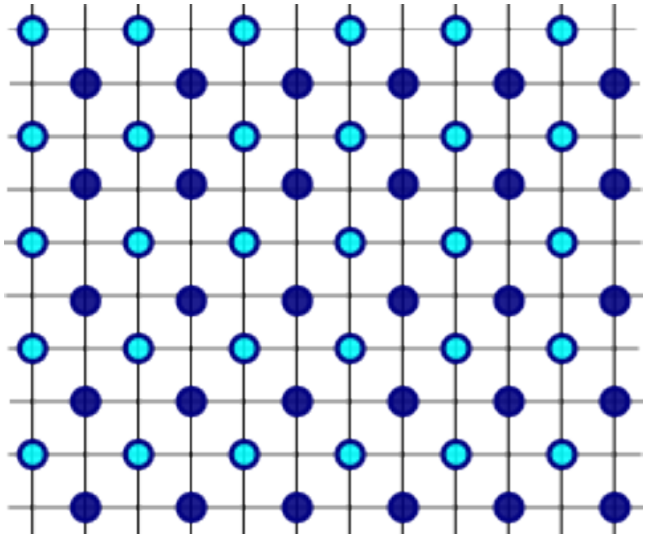
The pre-image $\rho^{-1}((a_1, a_2)) \subset \mathbb{Z}^2$ of $(a_1 \pmod 3, a_2 \pmod 3)$.

$$\rho : \mathbb{Z}^n \to \mathbb{Z}_q^n, \ \mathbf{x} \mapsto \rho(\mathbf{x}).$$

One may take any arbitrary subset S of $\mathbb{Z}_q^n$ and compute $\rho^{-1}(S)$, but what if S has a structure? how is it carried over to $\rho^{-1}(S)$?



The pre-image $\rho^{-1}(S) \subset \mathbb{Z}^2$ of $S = \{(a_1 \pmod 3), a_2 \pmod 3)), (a_1 + 1 \pmod 3), a_2 + 1 \pmod 3))\}$.



The pre-image $\rho^{-1}(C) \subset \mathbb{Z}^2$ of the linear binary code $C = \{(0,0), (1,1)\}$.

A well known connection between linear codes in $\mathbb{Z}_q^n$ and lattices: Given a subset $S \subset \mathbb{Z}_q^n$ then $\rho^{-1}(S)$ is a lattice in $\mathbb{R}^n$ if and only if S is a linear code in $\mathbb{Z}_q^n$.

A linear code C in $\mathbb{Z}_q^n$ is an additive subgroup of $\mathbb{Z}_q^n$.

A lattice $L \subset \mathbb{R}^n$ is an additive subgroup of $\mathbb{R}^n$ given by

$$L = \{uM,\ u \in \mathbb{Z}^m\}$$

for M a generator matrix, whose rows are $m \leq n$ linearly independent vectors.

Construction A

Let C be a linear code in $\mathbb{Z}_q^n$, the integers modulo a positive integer $q \geq 2$, where q is either prime or composite. Let $\rho : \mathbb{Z}^n \to \mathbb{Z}_q^n$ be the componentwise reduction modulo q. Then the lattice $\Lambda_C = \rho^{-1}(C)$ is said to have been obtained via *Construction A*.

- Sometimes $\Lambda_C = \frac{1}{\sqrt{q}}\rho^{-1}(C)$ is called Construction A.
- There are other constructions, Construction B,C,D...
- The lattice Λ_C is also known as a q-ary lattice or modulo q lattice (especially in the context of lattice-based cryptography).

✓ **Lattices**

Construction A

✓ **Coding Theory**

Cryptography

Let L be a lattice. Suppose d is the smallest distance between any two distinct points of L, the vectors in L with length d are the shortest vectors of L.

- The minimum norm of L, or the norm of L or the minimum of L, is the length of a shortest nonzero lattice vector of L:

$$N(L) = \min(L) = \inf_{\mathbf{x} \in L, \mathbf{x} \neq 0} N(\mathbf{x}),$$

  where the norm of $\mathbf{x}$ is $N(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} = (||\mathbf{x}||_2)^2$.
- For $L = \mathbb{Z}^n$, $N(L) = 1$.
- Let L be a lattice of rank n, then

$$\gamma(L) = \frac{N(L)}{\sqrt[n]{\det(L)}}, \ \det(L) = \det(G),$$

  which is invariant by scaling.

Consider the lattice L with generator and Gram matrices

$$M = \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{3} \end{bmatrix}, \; G = MM^T = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}.$$

Then $\det(L) = \det(G) = 12$.
The norm of $\mathbf{x} = \mathbf{u}M$ for $u = (u_1, u_2) \in \mathbb{Z}^2$ is

$$\begin{aligned} N(\mathbf{x}) &= \mathbf{u}M(\mathbf{u}M)^T \\ &= \mathbf{u}G\mathbf{u}^T \\ &= 4(u_1^2 + u_1 u_2 + u_2^2). \end{aligned}$$

Therefore, $N(\mathbf{x}) \geq 4$ for $\mathbf{x} \neq 0$ in L and $N(L) = 4$ (both basis vectors have norm 4).
Consequently

$$\gamma(L) = \frac{N(L)}{\sqrt{det(L)}} = \frac{4}{\sqrt{12}} = \frac{2}{\sqrt{3}}.$$

Hermite's constant (named after Charles Hermite (1822-1901)) is defined by

$$\gamma_n = \sup_{\dim(L)=n} \gamma(L) = \sup_{\dim(L)=n} \frac{N(L)}{\sqrt[n]{\det(L)}}.$$

An n-dimensional lattice L is a critical lattice if $\gamma(L) = \gamma_n$.

Known values of Hermite's constant:

n	1	2	3	4	5	6	7	8	24
γ_n	1	$\frac{2}{\sqrt{3}}$	$\sqrt[3]{2}$	$\sqrt{2}$	$\sqrt[5]{8}$	$\sqrt[6]{\frac{64}{3}}$	$\sqrt[7]{64}$	2	4

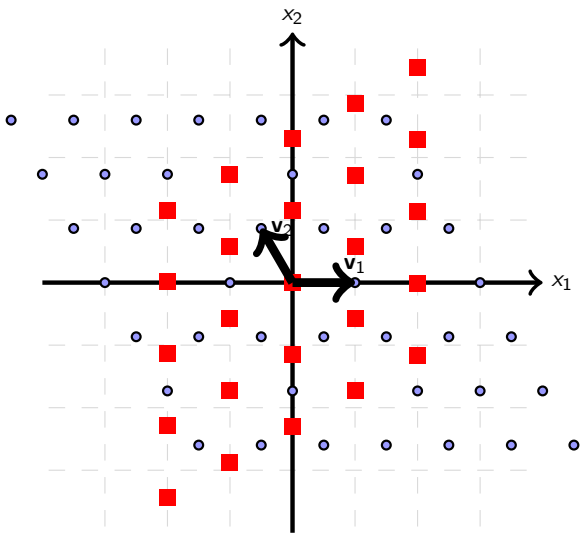
- A sublattice $L' \leq L$ of rank l is a lattice included in L, generated by l basis vectors.
- Set

$$d_l(L) = \min_{\substack{L' \leq L, \\ rank(L')=l}} \det(L'),\ 1 \leq l \leq n.$$

- We define

$$\begin{aligned} \gamma_{n,l}(L) &= d_l(L) \det(L)^{-l/n} \\ \gamma_{n,l} &= \sup_{rank(L)=n} \gamma_{n,l}(L). \end{aligned}$$

The constant $\gamma_{n,l}$ is called the Rankin constant, and when $l = 1$, it translates into the Hermite constant: $\gamma_{n,1} = \gamma_n$.



We define

$$\begin{aligned} \gamma'_{n,l}(L) &= \sqrt{d_l(L)d_l(L^*)} \\ \gamma'_{n,l} &= \sup_{rank(L)=n} \gamma'_{n,l}(L), \end{aligned}$$

where $L^* = \{\mathbf{y} \in \mathbb{R}^n,\ \mathbf{y} \cdot \mathbf{x} \in \mathbb{Z}$ for all $\mathbf{x} \in L\}$ denotes the dual lattice of L and

$$d_l(L) = \min_{\substack{L' \leq L, \\ rank(L')=l}} \det(L'),\ 1 \leq l \leq n.$$

The constant $\gamma'_{n,l}$ is sometimes called the Bergé-Martinet constant.

Determining $\gamma_{n,l}$ and $\gamma'_{n,l}$ also remains mostly open. [K. Sawatani, T. Watanabe, K. Okuda, "A note on the Hermite-Rankin constant"]

n	l	$\gamma_{n,l}$	L	$\gamma'_{n,l}$	L
2	1	$\frac{2}{\sqrt{3}}$	A_2	$\frac{2}{\sqrt{3}}$	
3	1	$\sqrt[3]{2}$	$A_3 \simeq D_3$	$\sqrt{3/2}$	D_3
4	1	$\sqrt{2}$	D_4	$\sqrt{2}$	D_4
4	2	$3/2$	D_4	$3/2$	D_4
5	1	$\sqrt[5]{8}$	D_5	$\sqrt{2}$	D_5
5	2	?			
6	1	$(\frac{64}{3})^{1/6}$	E_6	$\sqrt{8/3}$	
6	2	$3^{2/3}$	E_6	2	E_6
6	3				
7	1	$\sqrt[7]{64}$	E_7	$\sqrt{3}$	
7	2				
7	3				
8	1	2	E_8	2	E_8
8	2	3	E_8	3	E_8
8	3	4	E_8	4	E_8
8	4	4	E_8	4	E_8

As for bounds:

- It is known [N. Gama, N. Howgrave-Graham, H. Koy, P.Q. Nguyen, "Rankin's constant and blockwise lattice reduction", CRYPTO 2006] that

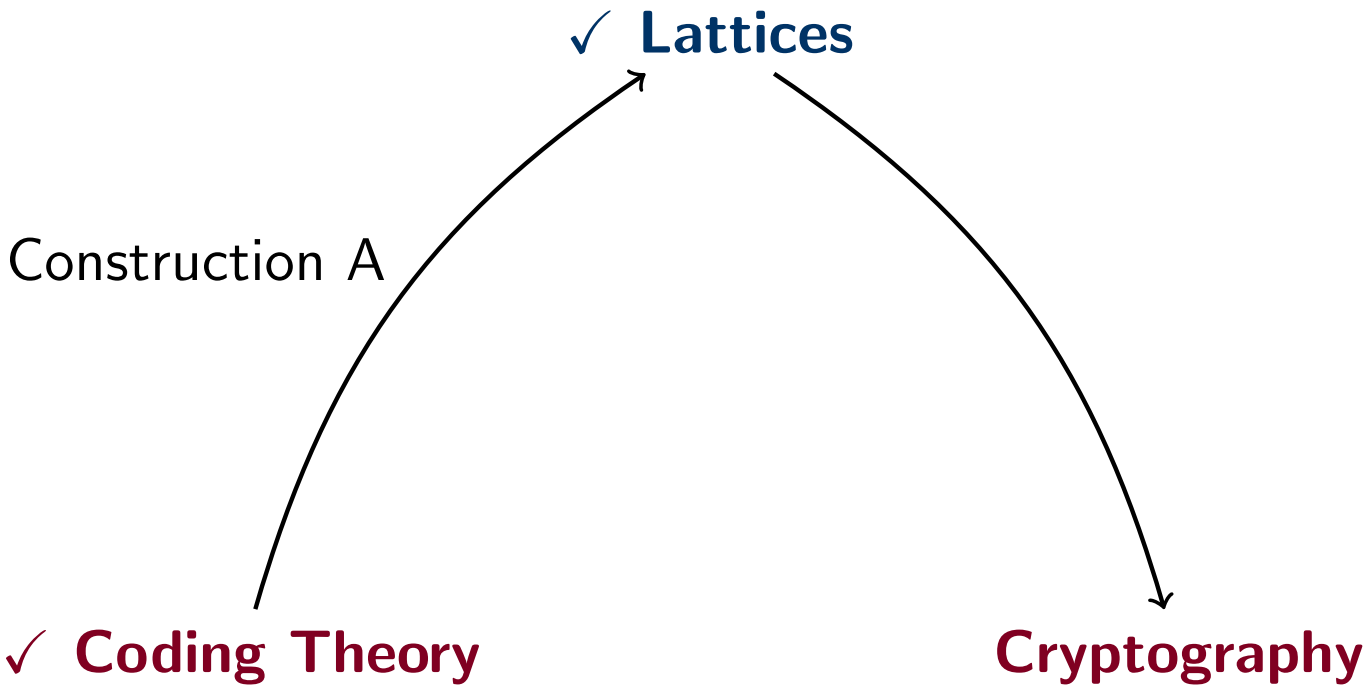
$$\left(\frac{k}{12}\right)^{k/2} \leq \gamma_{2k,k} \leq \left(1+\frac{k}{2}\right)^{k\ln 2+1/2}, \ k \geq 2,$$

- This result was improved, for $k \geq 5$, to

$$\frac{4}{\pi^2\sqrt{k}}\left(\frac{2k}{\pi e^{3/2}}\right)^{k/2} \leq \gamma_{2k,k} \leq e^9(0.0833)^{k/2}\left(\frac{4k-1}{17}\right)^{\frac{k}{4k-2}}(k-0.5)^{k\ln 2}$$

  in [J. Wen, X.W. Chang, "Sharper bounds on four lattice constants", DCC 2022].

- Bounds for arbitrarily large dimensions are of interest for lattice reductions, a topic that attracted a renewed interest in the context of lattice-based cryptography.


✓ Lattices
Construction A
✓ Coding Theory
Cryptography

A Reformulation of $\gamma_{n,l}(\Lambda_C)$

For C a linear code in $\mathbb{Z}_q^n$ and the lattice $\Lambda_C = \rho^{-1}(C)$, we have

$$\gamma_{n,l}(\Lambda_C) = d_l(\Lambda_C)\frac{|C|^{2l/n}}{q^{2l}}.$$

- This comes from a reformulation of the lattice determinant as a function of the code parameters (which follows from $|\Lambda_C/q\mathbb{Z}^n| = |C|$).

A First Bound on $\gamma_{n,l}(\Lambda_C)$

We have $d_l(\Lambda_C) \leq q^{2l}$ so

$$\gamma_{n,l}(\Lambda_C) \leq |C|^{2l/n}.$$

- We have
$$d_l(\Lambda_C) \leq q^{2l}$$
since the lattice $q\mathbb{Z}^n$ is always a sublattice of Λ_C (indeed $\rho(q\mathbb{Z}^n) = \mathbf{0}$ which is always a codeword of C since the code is linear), which always contains the sublattice $q\mathbb{Z}^l$ for $1 \leq l \leq n$, for which a Gram matrix is $q^2 I_l$ and $\det(q\mathbb{Z}^l) = q^{2l}$.

A First Sequence

For $C = \{(c_1, \ldots, c_{n-1}, \sum_{i=1}^{n-1} c_i),\ c_1, \ldots, c_{n-1} \in \mathbb{Z}_q\}$, Λ_C satisfies

$$\gamma_{n,1}(\Lambda_C) = \frac{2}{q^{2/n}}.$$

1. $\Lambda_C = D_n$ when $q = 2$.
2. for $l = 1$, $\gamma_{n,1}(D_n) \geq \gamma_{n,1}(\Lambda_C)$, meaning that the case $q = 2$ provides the best $\gamma_{n,1}(\Lambda_C)$ among choices of q.
3. $\gamma_{n,1}(D_n)$ is optimal for $n = 3, 4, 5$,
4. for $l = 2$ and $n \geq 3$, $d_2(\Lambda_C) = 3$,

$$\gamma_{n,2}(D_n) = \frac{3}{4^{2/n}} \geq \gamma_{n,2}(\Lambda_C) = \frac{3}{q^{2/n}}$$

and $\gamma_{n,2}(D_n)$ is optimal for $n = 4$ while it serves as a benchmark for other values of n, namely

$$\gamma_{n,2} \geq \frac{3}{4^{2/n}},\ n \geq 3.$$

For $q = p$ prime, and

$$\begin{bmatrix} \mathbf{I}_k & A \end{bmatrix},$$

the generator matrix of C in systematic form, then a generator matrix of Λ_C is

$$\begin{bmatrix} \mathbf{I}_k & A \\ \mathbf{0} & q\mathbf{I}_{n-k} \end{bmatrix}.$$

For $n \geq 2$, $C = \{(a_1, \ldots, a_{n-1}, \sum_{i=1}^{n-1} a_i),\ a_1, \ldots, a_{n-1} \in \mathbb{F}_2\}$ has length n and dimension $n-1$. A systematic generator is

$$\begin{bmatrix} \mathbf{I}_{n-1} & \mathbf{1} \end{bmatrix}.$$

A generator matrix for Λ_C is thus

$$\begin{bmatrix} \mathbf{I}_{n-1} & \mathbf{1} \\ \mathbf{0} & 2 \end{bmatrix}.$$

We have just constructed the lattice

$$D_n = \{(x_1, \ldots, x_n),\ \sum_{i=1}^{n} x_i \text{ is even}\}.$$

The Cases $\gamma_{5,2}$ and $\gamma_{7,2}$

We have

$$1.723 \approx \frac{3}{4^{2/5}} \leq \gamma_{5,2} \leq 2, \; 2.0189 \approx \frac{3}{4^{2/7}} \leq \gamma_{7,2} \leq 2^{5/3} \approx 3.1748.$$

- The lower bound follows from above, namely $\gamma_{n,2} \geq \frac{3}{4^{2/n}}$.
- The upper bound follows from the known inequality $(\gamma_{n,l})^{n-2l} \leq (\gamma_{n-l,l})^{n-l}$.
- For $n = 5$, we have

$$\gamma_{5,2} \leq (\gamma_{3,2})^3 = 2$$

while for $n = 7$

$$(\gamma_{7,2})^3 \leq (\gamma_{5,2})^5 \leq 2^5.$$

Let $\mathcal{R}(r, m)$ denote the binary Reed-Mueller code of order r and length 2^m. A recursive formula for a generator matrix of $\mathcal{R}(r, m)$ is

$$\mathbf{B}_{r,m} = \begin{bmatrix} \mathbf{B}_{r,m-1} & \mathbf{B}_{r,m-1} \\ \mathbf{0} & \mathbf{B}_{r-1,m-1} \end{bmatrix},$$

with $\mathbf{B}_{0,i} = \mathbf{1}_{1,2^i}$ and $\mathbf{B}_{i,i} = I_{2^i}$ for $i \leq m$.

- The dimension k of this code is $k = \sum_{i=0}^{r} \binom{m}{i}$.
- We have $\mathcal{R}(i, m) \subseteq \mathcal{R}(j, m)$ for $i \leq j$.
- Given the lattice $\Lambda_{\mathcal{R}(j,m)}$, a natural sublattice is generated by $\mathbf{B}_{i,m}$, a generator matrix of $\mathcal{R}(i, m)$, which we will denote by $\Lambda(\mathbf{B}_{i,m})$.

The Case $r = 1$

1. For $l = 1$,

$$\gamma_{n,1}(\Lambda_{\mathcal{R}(1,m)}) = \begin{cases} \frac{2^{3/2}}{2} = \sqrt{2} & m = 2 \\ 2^{2(m+1)/2^m} & m \geq 3, \end{cases}$$

   which is optimal for $m = 2, 3$ (that is in dimension 4 and 8).
2. For $l = 2$, there exists a sublattice $\Lambda(\mathbf{B})$ such that

$$\gamma(\Lambda_{\mathcal{R}(1,m)}, \Lambda(\mathbf{B})) = \begin{cases} 3 & m = 3 \ (n = 8) \\ \frac{1}{2\sqrt{2}} & m = 4 \ (n = 16) \end{cases}$$

   and $\gamma_{8,2}(\Lambda_{\mathcal{R}(1,3)}) = 3$, which is optimal in dimension 8.
3. For $l \geq 3$, there exists a sublattice $\Lambda(\mathbf{B})$ such that

$$\gamma(\Lambda_{\mathcal{R}(1,m)}, \Lambda(\mathbf{B})) = \frac{4 \cdot 2^{(m-2)l}}{\left(\frac{2^{2^m}}{2^{m+1}}\right)^{2l/n}}$$

   for $m \leq \frac{2l-2}{l} + 2$. This quantity simplifies to 4 when $m = 3$ and $l = 3, 4$.

The Case $r = m - 1$

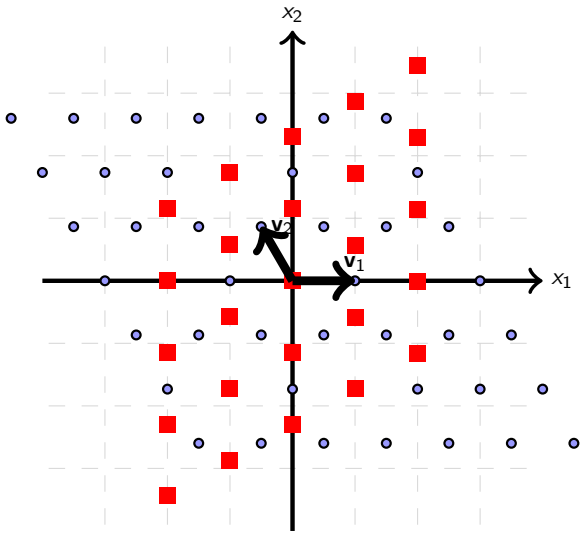
1. For $l = 1$,
$$\gamma_{n,1}(\Lambda_{\mathcal{R}(m-1,m)}) = \frac{2}{2^{2/2^m}}.$$
2. For $l \geq 2$,
$$\gamma_{n,l}(\Lambda_{\mathcal{R}(m-1,m)}) \leq \begin{cases} 3 \cdot 2^{2(m-2)-4/2^m} & l = 2 \\ 4 \cdot 2^{2(m-2)-2l/2^m} & l \geq 3. \end{cases}$$

About $\gamma'_{n,l}(\Lambda_C)$

- We recall that $\gamma'_{n,l}(L) = \sqrt{d_l(L)d_l(L^*)}$ where L^* is the dual lattice of L.
- For speaking of duality, it is often preferred to work with $\Lambda_C = \frac{1}{\sqrt{q}}\rho^{-1}(C)$ (we will see why next).
- The results hold for $\mathbb{F}_q$ (though we consider $q = p$).

$C^\perp = \{\mathbf{v} \in \mathbb{F}_q^n,\ \mathbf{c}\cdot\mathbf{v} = \mathbf{0}$ for all $\mathbf{c} \in C\}$	$L^* = \{\mathbf{y} \in \mathbb{R}^n,\ \mathbf{y}\cdot\mathbf{x} \in \mathbb{Z}$ for all $\mathbf{x} \in L\}$

We have $\frac{1}{\sqrt{q}}\Lambda_{C^\perp} = (\frac{1}{\sqrt{q}}\Lambda_C)^*$, or equivalently $\frac{1}{q}\Lambda_{C^\perp} = \Lambda_C^*$.



- Then $C = C^\perp$ if and only if $\frac{1}{\sqrt{q}}\Lambda_C = (\frac{1}{\sqrt{q}}\Lambda_C)^*$ if and only if $\frac{1}{\sqrt{q}}\Lambda_C$ is unimodular.
- Normalizing gives a determinant of 1.

A Sequence

For $C = \{(c_1, \ldots, c_{n-1}, \sum_{i=1}^{n-1} c_i),\ c_1, \ldots, c_{n-1} \in \mathbb{Z}_q\}$, we have:

1. $\gamma'_{n,1}(\Lambda_C) = \frac{1}{q}\sqrt{2\min(n, q^2)}$. In particular, if $n \geq q^2$, then $\gamma'_{n,1}(\Lambda_C) = \sqrt{2}$ so we have $\gamma'_{n,1} \geq \sqrt{2}$ when $n \geq 4$.
2. For $q = 2$, $l = 1$, we have $\gamma'_{n,1}(\Lambda_C) = \frac{1}{2}\sqrt{2\min(n, 4)}$. In particular,

$$\gamma'_{2,1}(\Lambda_C) = 1,\ \gamma'_{3,1}(\Lambda_C) = \sqrt{3/2},\ \gamma'_{4,1}(\Lambda_C) = \gamma'_{5,1}(\Lambda_C) = \sqrt{2}$$

   and $\gamma'_{n,1}(\Lambda_C)$ is optimal when $n = 3, 4, 5$.
3. For $n \geq 3$, we have $\gamma'_{n,2}(\Lambda_C) = \frac{1}{q}\sqrt{3\min(q^2, n-1)}$, so for $q = 2$ and $n = 4$, $\gamma'_{4,2}(\Lambda_C) = 3/2$ is optimal. Furthermore, $\gamma'_{n,2} \geq \sqrt{3}$ for $n \geq 5$.

The Cases $\gamma'_{5,2}$ and $\gamma'_{7,2}$

We have

$$1.7321 \approx \sqrt{3} \leq \gamma'_{5,2} \leq 2, \; 1.7321 \approx \sqrt{3} \leq \gamma'_{7,2} \leq \frac{8}{3} \approx 2.6667.$$

- The lower bound follows from above proposition, namely $\gamma'_{n,2} \geq \sqrt{3}$.
- The upper follows from the known inequality $\gamma'_{n,2l} \leq (\gamma'_{n-l,l})^2$.
- For $n = 5$, we have

$$\gamma'_{5,2} \leq (\gamma'_{4,1})^2 = 2$$

while for $n = 7$

$$\gamma'_{7,2} \leq (\gamma_{6,1})^2 \leq 8/3.$$

More on $\gamma_{5,2}$

We have $\gamma_{n,2} \geq \sqrt{3}$ for $n \geq 5$ so

$$1.7321 \leq \gamma_{5,2} \leq 2.$$

- We use $\gamma'_{n,l} \leq \gamma_{n,l}$ for the lower bounds.

Natural correspondence between codes and lattices:

dual code	dual lattice
self-dual code	unimodular lattice
weight enumerator	theta series
?	Rankin constants

- Known lattices:
  - Give known optimal constructions in small dimensions but still valid bounds for large dimensions.
- Bounds on $\gamma_{n,2}$ and $\gamma'_{n,2}$.
  - Due to the ability to provide a lower bound, which give new bounds for $n = 5, 7$.
- Bounds on families of $\gamma_{n,l}(\Lambda_C)$.

Open Questions

- $\gamma_{5,2}, \gamma'_{5,2}$ and $\gamma_{7,2}, \gamma'_{7,2}$?
- Other families of codes?

Thank You

For more: "About the Rankin and Bergé-Martinet Constants from a Coding Theory View Point", F. O., Shengwei Liu, Hongwei Liu.